

Module 2: Agentic AI

Architectures and Design Patterns

Demo-1

Demo-1: Designing an AI Agent Architecture

Use Case: AI Patient Support Agent for MediBo

To enhance patient care and lessen the burden on healthcare personnel, MediBo, a healthcare company, is implementing an AI-powered virtual healthcare assistant. This agent will help independently manage regular support duties, enable smart triaging, and effortlessly connect with electronic health systems. Key objectives that the agent might deliver:

1. Addressing questions about appointments and availability.
2. Responding to commonly asked medical inquiries.
3. Overseeing reminders for medications and their refills.
4. Starting pre-consultation symptom assessment.
5. Intensifying urgent symptoms or emergency demands.
6. Gaining insights from exchanges to enhance service customization.

The point that needs to be remembered is that all the operations must comply with policies that have been implemented to safeguard the privacy and security of the medical records and other patient personal data interacting with the application, data protection laws, and securely integrate with electronic health records (EHR), scheduling systems, and pharmacy APIs.

Step 1: Defining Agent Capabilities

To build a meaningful agent, begin by clearly outlining its roles:

1. What patient tasks will the agent fully automate?
2. When is clinician or nurse intervention needed?
3. Which external systems and healthcare APIs (EHR, lab results, pharmacy) must it interact with?
4. How frequently does the context change (e.g., medication plans, appointment protocols)?

Step 2: Interpreting Capabilities to Functionalities

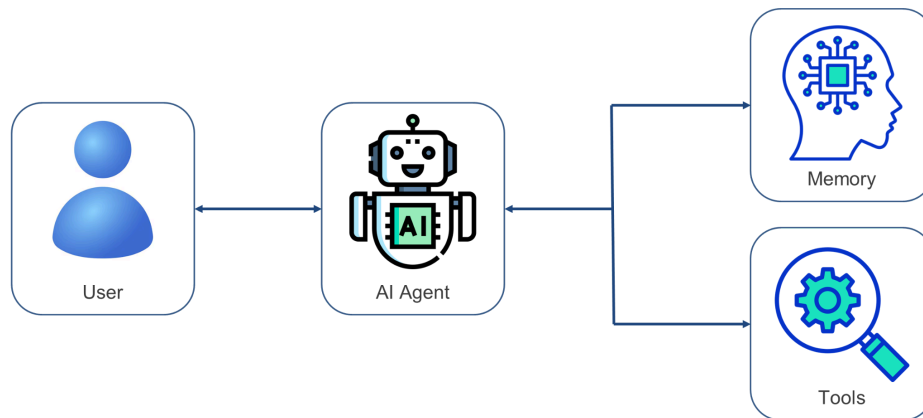
Translating Capabilities to application Functionalities: -

Capabilty	Functionality
Comprehend patient inquiries through text or voice	Natural Language Processing (NLP) for inquiries regarding symptoms, medications, or scheduling.
Connect with EHR and pharmacy APIs	Capability to retrieve/update appointment, prescription, and test records.
Carry out smart triage	Classification based on symptoms for urgent, non-urgent, and emergency indicators.
Study previous instances	Identify recurring issues or actions to tailor engagements.
Guarantee adherence & confidentiality	Management of encrypted data, user verification, consent monitoring.

Step 3: Selecting Type of Architecture

1. Single Agent Architecture

One AI agent incorporates all functionalities—perception, cognition, action, and learning—into a single cohesive system. Incase of this usecase, it can be considered ideal for pilot projects where a chatbot responds to fundamental healthcare related inquiries.



Advantage:

- Easy to set up – Ideal for pilot programs like appointment booking or symptom checks.
- Rapid deployment – Minimal integration needed, faster time to value.
- Low infrastructure complexity – Simplified maintenance with fewer moving parts.



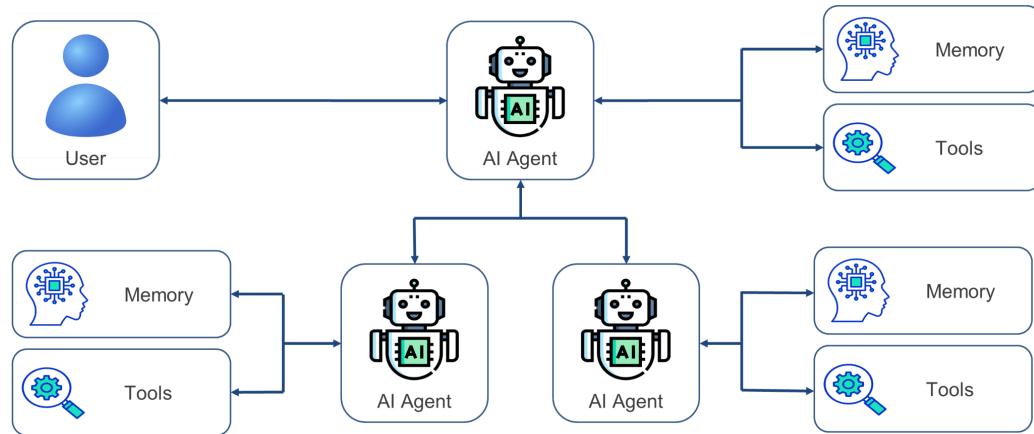
Disadvantage:

- Limited scalability – Has difficulty handling various workflows (e.g., billing versus diagnostics).
- Tight coupling – Any modification impacts the whole system; challenging to sustain in the long run.
- Debugging and upgrades carry risks – Issues or newly added features may lead to widespread system disturbance.

2. Multi-Agent Architecture

Numerous specialized agents function autonomously and collaborate to manage intricate workflows. Every agent possesses specific, restricted abilities (e.g., one for monitoring, one for grievances, one for modifications

to any booking). In this scenario, it is perfect for large-scale patient support, where agents engage with various backend systems and must operate simultaneously.



Advantage:

- Adapts effectively to various business sectors.
- Enables simultaneous execution of tasks.
- Greater fault tolerance—an issue in one agent doesn't compromise the entire system.
- Simpler to disseminate among teams and systems

Disadvantage:

- Needs protocols for communication between agents.
- Increased coordination and orchestration costs.
- Identifying issues in distributed interactions can be challenging.

3. Hybrid Architecture

One central coordinating agent serves as the hub, yet delegates certain tasks (such as perception, learning, security, analytics) to modular services operating as microservices or function-oriented APIs. In this scenario, it is strongly advised as it fosters a quality intelligent healthcare agent

featuring functionalities such as multilingual support, real-time tracking, personalization, and ongoing learning.



Advantage:

- Offers balance between maintainability and autonomy.
- Modules can be independently deployed, scaled, and versioned.
- Integrates effectively with EHR systems, pharmacy solutions, and cloud services.
- Modules can undergo versioning and enhancements separately.
- Simpler to identify and oversee critical processes (for instance, HIPAA logging).

Disadvantage:

- Requires robust API management, message queues, or service mesh.
- Risks related to data consistency and latency can occur which means communication between modules needs to be carefully controlled.
- Necessitates robust CI/CD, monitoring, and rollback plans.

Step 4: Identify Agentic Modules

Objective: Create a flexible Agentic AI structure designed to enhance healthcare communication and automate routine patient activities. for MediBo.



Module	Use case Functionality
Perception Module	Comprehends multimodal inputs (text/voice), interprets patient inquiries such as “What time is my appointment?” or “Is it necessary to take this with food?” utilizing a chatbot or voice interface.
Cognitive Module	Understands purpose (like, arranging appointments, medication management, prioritization), utilizes care procedures, reviews past records for context-driven insights.
Action Module	Schedules appointments, starts prescription renewals, notifies nurses during emergencies, sends follow-up communications.
Learning Module	Modifies responses according to patient history, enhances triage models utilizing information from previous symptom assessments, and upgrades suggestions.
Collaboration Module	Collaborates with healthcare personnel or clinicians in complex situations; engages with external systems (laboratories, insurance companies).
Security Module	Verifies patient identity, records actions for auditing, ensures all data transfers adhere to HIPAA and internal security regulations.

Scenario: Symptom-Based Triage Request

A patient communicates via a message using their healthcare provider's platform, that is the MediBo application. Suppose the user puts in a query, "I experience tightness in my chest and difficulty in breathing."

This is a practical example that showcases the significance of smart, adaptive, and secure automation in the healthcare sector. It emphasizes how AI agents can support critical triage scenarios, particularly during off-hours, high demand periods, or in remote locations where prompt clinical staff may not be present.

Spotting Problem: Recognizing and Reacting to a High-Risk Symptom

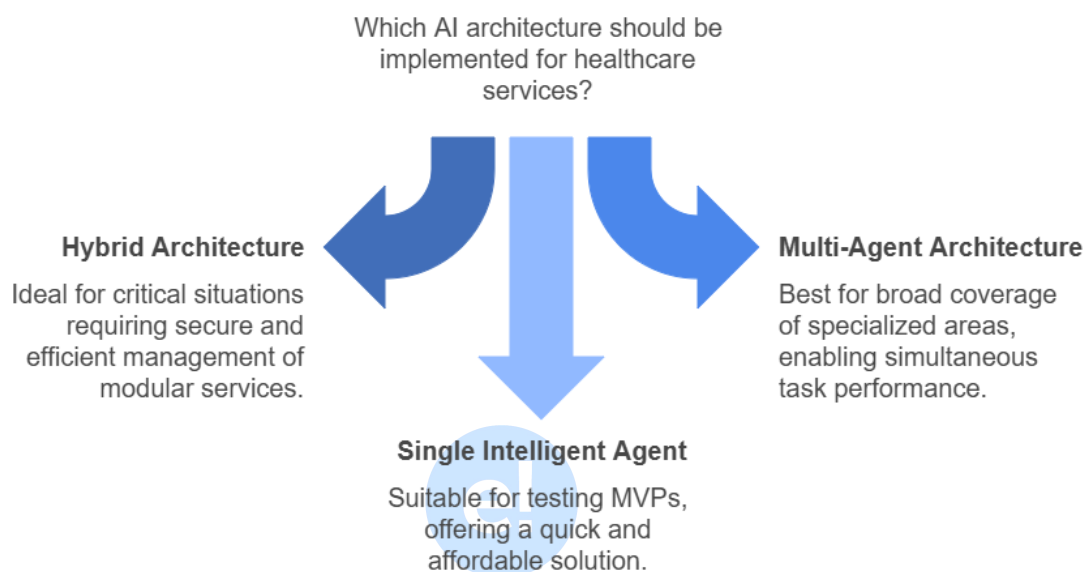
The difficulty in this situation is in acknowledging the seriousness of the symptoms and reacting suitably without hesitation. Without AI triage, the message might linger in a queue or be noticed too late by a nurse or clinician. These signs, chest pressure and difficulty breathing, are warning signals that may suggest a heart attack, asthma episode, or other critical emergencies. Conventional systems might overlook the immediacy or depend solely on human intervention, heightening risk.

Solution: Modular AI Agent Responds in Real Time

Employing a modular Agentic AI framework, the system efficiently analyzes the message via its Perception Module, utilizing natural language understanding (NLU) to identify essential symptoms. The Cognitive Module assesses the situation and utilizes triage guidelines or a symptom knowledge graph to determine the level of urgency. Subsequently, the Action Module issues a notification to the on-call clinical staff and may also provide the patient with urgent advice, like: "Contact emergency services if you're by yourself or feeling worse."

When needed, the Collaboration Module elevates issues to a human nurse, while the Security Module guarantees that data is encrypted and access-controlled in compliance with HIPAA/GDPR regulations. In the meantime, the Learning Module documents the complete interaction and patient result, enhancing future triage precision via feedback.

Building on the Basis of Architecture According to the Use Case



1. In critical situations such as symptom assessment (e.g., tightness in the chest and difficulty breathing), it is advisable to start with a Hybrid Architecture. What this does is that it allows a central intelligent agent to manage various modular services like symptom assessment, EHR access, escalation processes, and data protection. It enables the system to securely connect with hospital resources (such as patient records, identity checks, and scheduling systems), guaranteeing compliance, efficiency, and clinical precision. The modularity also allows teams to implement updates to specific components—like enhancing the triage logic—without needing to revamp the whole system.
2. Suppose the healthcare agent broadens its coverage to include more specialized areas, like mental health assistance, chronic disease management, insurance claims processing, or prescription logistics. A

Multi-Agent Architecture becomes optimal choice in a situation like this. So, single agents can manage particular services such as triage, medication refills, lab result follow-ups, or insurance confirmations. This encourages simultaneous task performance, specialized knowledge in specific areas, and shared dependability. For instance, a billing representative wouldn't disrupt a symptom checker, but both would still contribute to a unified patient experience.

3. In case what you want is testing a minimum viable product (MVP), like a chatbot that schedules appointments or responds to common medical inquiries, then a Single Intelligent Agent is a quick and affordable starting solution to tackle this. It integrates all capabilities (perception, decision-making, action, learning) into one deployable service. Although narrow in focus, this arrangement can be useful in outpatient environments or as an aid for front-desk personnel, and it provides a foundation for future modular growth.

Let's Understand the End-to-End Workflow on the Basis of Scenario

1. User starts a request through the MediBo chat interface or platforms to inquire about health symptoms.
2. **Perception Module** transforms input (multimodal) into a structured query. The system obtains the patient's message through chat, voice, or application input.
 - a. Utilizes NLP and medical language models to analyze input and identify intent along with clinical terms such as "chest tightness" and "shortness of breath."
 - b. Identifies urgency by evaluating tone, symptom severity, and possible risk signals.
3. **Cognitive Module**
 - a. Analyzes the organized input to recognize that the patient is indicating possible cardiac or respiratory symptoms.

- b. Utilizes a medical knowledge graph to identify high-risk patterns and assesses if symptoms may suggest an emergency (e.g., possible heart attack).
- c. Determines if the case should be directed for self-care advice, teleconsultation, or urgent escalation.

4. Action Module, when the symptoms reach a crucial threshold, the agent:

- a. Activates a notification for on-call medical personnel.
- b. Books an immediate video consultation or recommends visiting the closest emergency room.
- c. Delivers a location-specific emergency suggestion through incorporated health services or ambulance APIs.
- d. If not urgent, it might advise home care measures or suggest a visit to a primary care provider.

5. Learning Module

- a. Records the complete interaction to enhance triage models according to clinical follow-up results.
- b. Identifies ambiguous or novel symptom patterns for human evaluation, aiding in enhancing future classification precision.
- c. Integrates user input (e.g., "I was taken to ER") to modify upcoming intent forecasts and risk levels.

6. Collaboration Module

- a. If the circumstances are unclear or emotionally heightened (e.g., an anxious tone is recognized), the agent smoothly transitions to a nurse or telehealth coordinator.
- b. Facilitates coordination among multiple agents (e.g., involving scheduling, billing, or pharmacy agents when necessary).
- c. Delivers transcripts or summaries to human personnel, ensuring continuity of care.

7. Security Module

- a. Confirms patient identity through secure verification methods (e.g., two-factor authentication, biometrics).
- b. Verifies that all communications, records, and activities align with the different policies and regulations it must follow.
- c. Records decisions and data access activities for accountability and auditing purposes.

